IIMSTC INTERSHIP DOCUMENT COHORT 8
(Documentation of 2nd Mini Project)

Date: 06/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 10

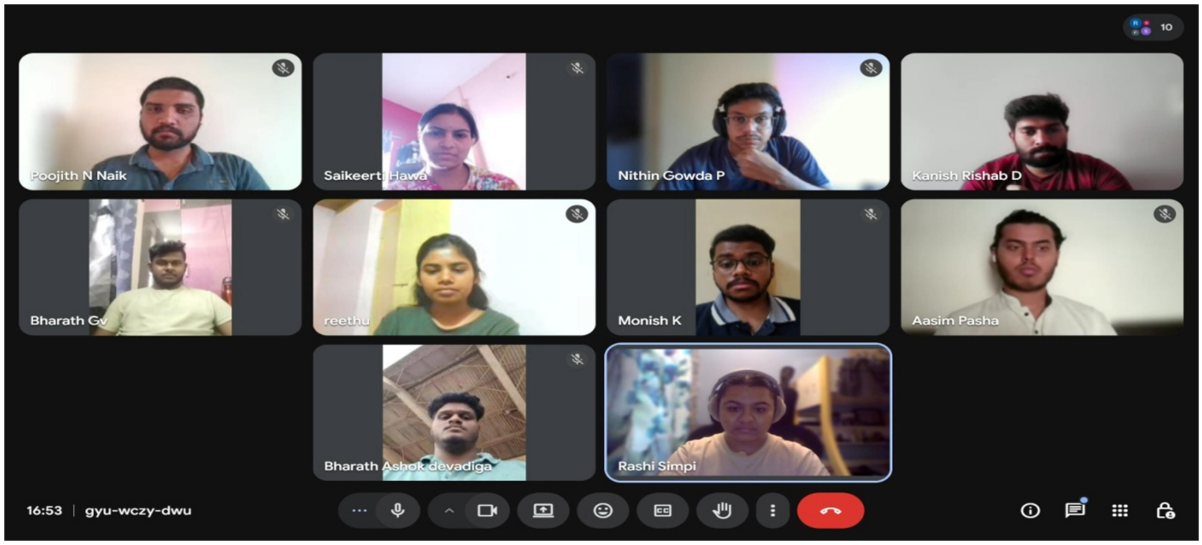
Team Members Absent: 5 (Mohammed Bilal Raza S H, Darshan Reddy G S, Ranjan, Mohammed Sameer, Khadija Tul Kubra)

Objective of the day: To initiate the **Poaching Threat Detection** project by understanding the problem statement, defining the project workflow, assigning responsibilities among team members, and forming dedicated teams for each stage of the machine learning pipeline.

Work Completed: During today's session, we held a discussion regarding the project titled **"Poaching Threat Detection"**, which focuses on identifying illegal activities such as poaching or logging in protected forests using acoustic sensor data. The overall workflow of the project was discussed, including stages such as data preprocessing, feature extraction, classifier model development, and model evaluation. To ensure smooth progress and effective teamwork, the cohort was divided into four specialized teams, each responsible for a specific phase of the project.

Team 1 was assigned with **Data Preprocessing** (Kanish Rishab D, Nithin Gowda P, Monish K). **Team 2** was assigned **Feature Extraction** (Aasim Pasha, Ranjan, Bharath G V, Khadija Tul Kurba). **Team 3** was assigned **Classifier Models** (Saikeerti Hawa, Darshan Reddy S, Mohammed Sameer, Mohammed Bilal Raza). **Team 4** was assigned **Model Evaluation** (Rashi Simpi, Poojith N Naik, Bharath Ashok Devadiga, Reethu G S). After the role allocation, each team began understanding their respective tasks and responsibilities to contribute effectively to the development of the project.

Attachments:



IIMSTC INTERSHIP DOCUMENT COHORT 8
(Documentation of 2nd Mini Project)

Date: 09/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 14

Team Members Absent: 1 (Mohammed Bilal Raza)

Objective of the day: To carry out the data collection, preprocessing, and labeling stages of the **Poaching Threat Detection** project by preparing the dataset in a structured format suitable for further machine learning tasks.

Work Completed: During today's work, the FSC22 audio dataset required for the project was downloaded from Kaggle and its structure was examined to understand the distribution of files and sound classes. The dataset contains 2025 audio files across 27 different sound classes stored in a single folder. The metadata file was inspected to identify the columns linking each audio file to its corresponding class label. A verification step was also performed to ensure that all audio files could be successfully loaded without corruption. Additionally, an organized project folder structure was prepared so that the dataset and processed outputs could be stored systematically for further stages.

Further processing was carried out on the raw audio data to convert it into a clean and standardized format suitable for machine learning models. Each audio file was converted to a mono channel with a consistent sampling rate. Silence present at the beginning and end of the recordings was removed to eliminate unnecessary segments. The duration of each audio clip was standardized to 5 seconds by trimming longer clips and padding shorter ones. After preprocessing, the processed audio files were organized into two main categories representing potential threat sounds and normal wildlife sounds.

In the next stage, the sound classes were analyzed to determine which classes represent illegal human activities in forest environments and which correspond to natural background sounds. Based on this analysis, specific classes related to activities such as machinery sounds or cutting tools were categorized as threat sounds, while the remaining classes were treated as wildlife sounds. A binary labeling approach was implemented to classify the dataset accordingly. Finally, a structured metadata file was generated containing details such as filename, original class label, binary label, encoded label, sample rate, duration, and file path, which will support the next phase of feature extraction and model development.

Doubts/Problems encountered during the Session: During the dataset preparation process, some difficulties were faced in understanding the dataset structure since all audio files were stored in a single folder. Identifying the correct metadata columns linking audio files to their classes also required careful verification. Additionally, handling variations in audio length and deciding the correct categorization of threat and wildlife sounds required discussion and adjustments during preprocessing.

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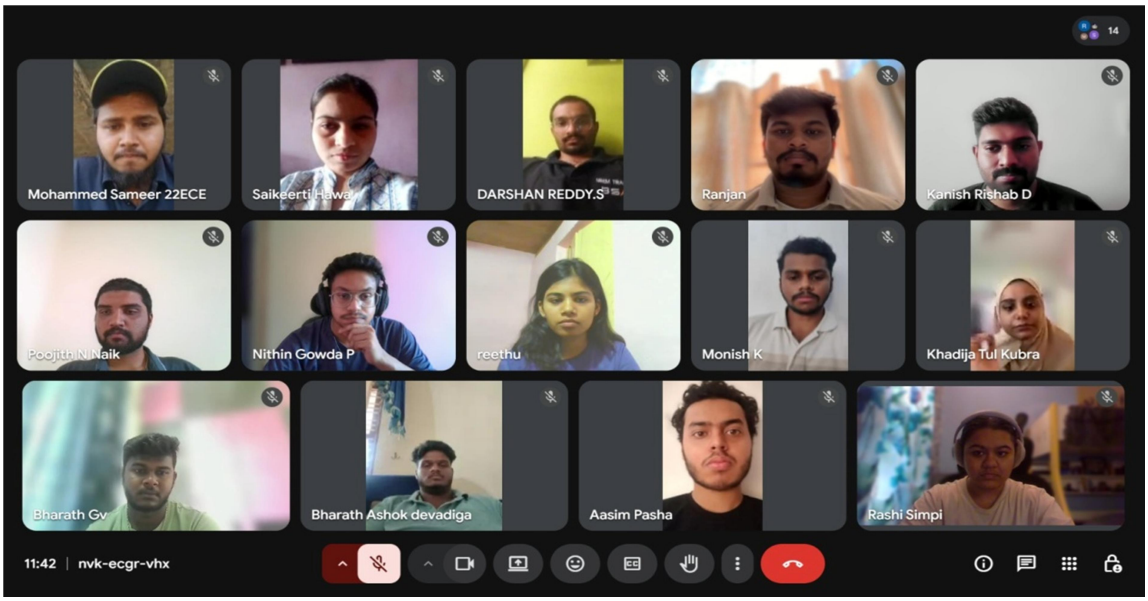

IIMSTC INTERSHIP DOCUMENT COHORT 8
(Documentation of 2nd Mini Project)

Date: 10/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 12

Team Members Absent: 3 (Mohammed Bilal Raza, Bharath Ashok Devadiga, Mohammed Sammer)

Objective of the day: To perform the **feature extraction stage** of the Poaching Threat Detection project by converting the processed audio data into numerical features that can be used for machine learning model training and analysis.

Work Completed: The processed audio files and metadata generated in the previous stage were first verified and prepared for the feature extraction process. The metadata file was loaded and the directory containing the processed audio files was indexed to efficiently locate each `.wav` file. A mapping between filenames and file paths was created to streamline the pipeline and reduce repeated file searches. A validation step was also performed to ensure that all audio files listed in the metadata could be successfully loaded without errors.

Next, the core feature extraction process was implemented using audio analysis techniques. Each processed audio file was analyzed to extract multiple spectral and temporal features that represent the sound signals numerically. Features such as MFCC, delta and delta-delta coefficients, Mel spectrogram, chroma features, spectral centroid, spectral bandwidth, spectral rolloff, spectral contrast, zero-crossing rate, and RMS energy were computed. Statistical summaries such as mean and standard deviation were calculated to produce a consistent numerical representation for every audio sample.

After feature extraction, the pipeline was executed across the entire dataset by iterating through all audio samples and collecting the extracted features. The resulting feature values were combined with relevant metadata such as filename and class labels. These features were then organized into a structured dataset where each row represents an audio sample and each column represents a specific feature. Finally, the dataset was exported as a CSV file that will be used for the next stage of machine learning model training and evaluation.

Doubts/Problems encountered during the Session: During the feature extraction stage, some challenges were encountered while ensuring that all audio files were properly linked with their metadata entries. Processing a large number of audio files also required careful validation to avoid missing or corrupted files. Additionally, selecting appropriate audio features that effectively represent different sound patterns required discussion and verification to ensure reliable input for the machine learning models.

Attachments:

12
DARSHAN REDDY.S
Kanish Rishab D
Saikeerti Hawa
Poojith N Naik
Ranjan
Aasim Pasha
Nithin Gowda P
Monish K
Bharath Gv
Khadija Tul Kubra
reethu
Rashi Simpi
14:23 | rtp-pwjy-wre

IIMSTC INTERSHIP DOCUMENT COHORT 8
(Documentation of 2nd Mini Project)

Date: 11/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 11

Team Members Absent: 4 (Mohammed Bilal Raza, Bharath Ashok Devadiga, Mohammed Sammer, Khadija Tul Kubra)

Objective of the day: To implement and evaluate different machine learning classification models for the Poaching Threat Detection project by preparing the dataset, training models, and analyzing their performance using evaluation metrics.

Work Completed: The dataset was first prepared for model training by performing the necessary preprocessing steps. The extracted feature dataset was divided into training and testing sets using an 80/20 split to ensure unbiased model evaluation. Feature normalization was performed using standard scaling techniques to remove variations in feature scales and improve model performance. Additionally, initial exploratory analysis was conducted using visualization techniques such as correlation heatmaps to understand relationships between features and identify possible multicollinearity that could influence model selection.

Following the preprocessing stage, a Random Forest classification model was developed using an ensemble approach that combines multiple decision trees to improve prediction accuracy and reduce variance. The model parameters were tuned to achieve optimal performance, and detailed classification reports were generated including metrics such as precision, recall, and F1-score for each class along with macro and micro averages. These evaluation metrics helped in understanding the model's ability to correctly identify different sound categories.

In addition, other classification algorithms were implemented to compare model performance. A K-Nearest Neighbors model was developed using Euclidean distance and cross-validation techniques to determine the optimal number of neighbors for prediction. An SVM classifier with a radial basis function kernel was also implemented to handle non-linear decision boundaries. Hyperparameter tuning techniques were applied to optimize model parameters, and the models were evaluated using confusion matrices and prediction results to analyze classification performance and identify misclassification patterns.

Doubts/Problems encountered during the Session: During the model development stage, some challenges were encountered in selecting appropriate hyperparameters for different algorithms to achieve optimal performance. Handling feature scaling and ensuring that all models received properly normalized input data also required careful verification. Additionally, interpreting confusion matrices and evaluation metrics for multiple models required analysis to determine which model performs better for the classification task.

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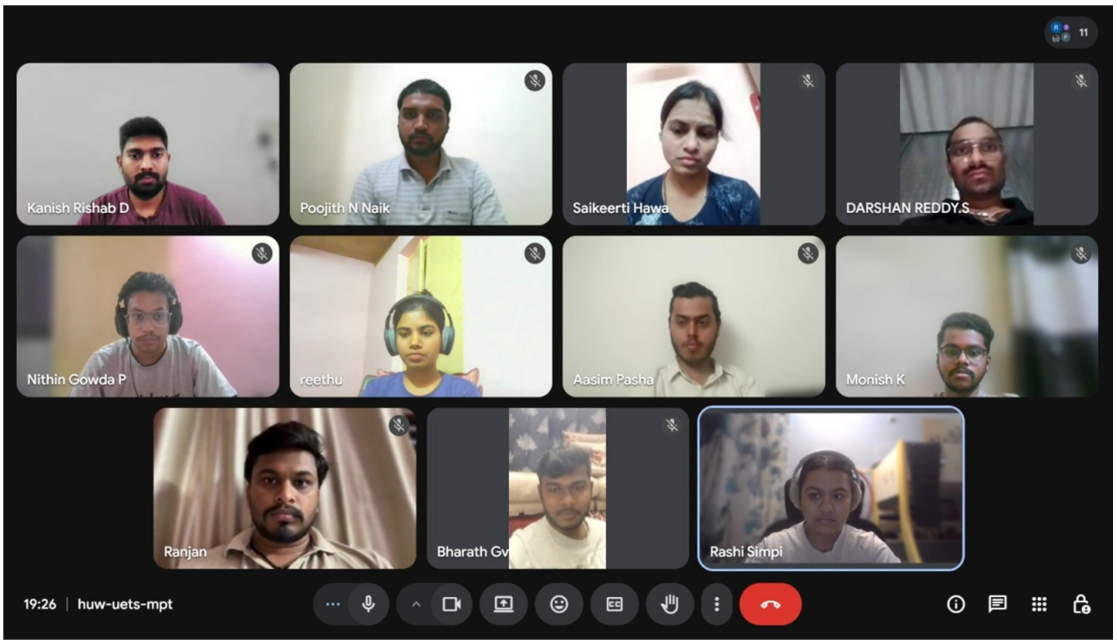


IIMSTC INTERSHIP DOCUMENT COHORT 8
(Documentation of 2nd Mini Project)

Date: 12/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 12

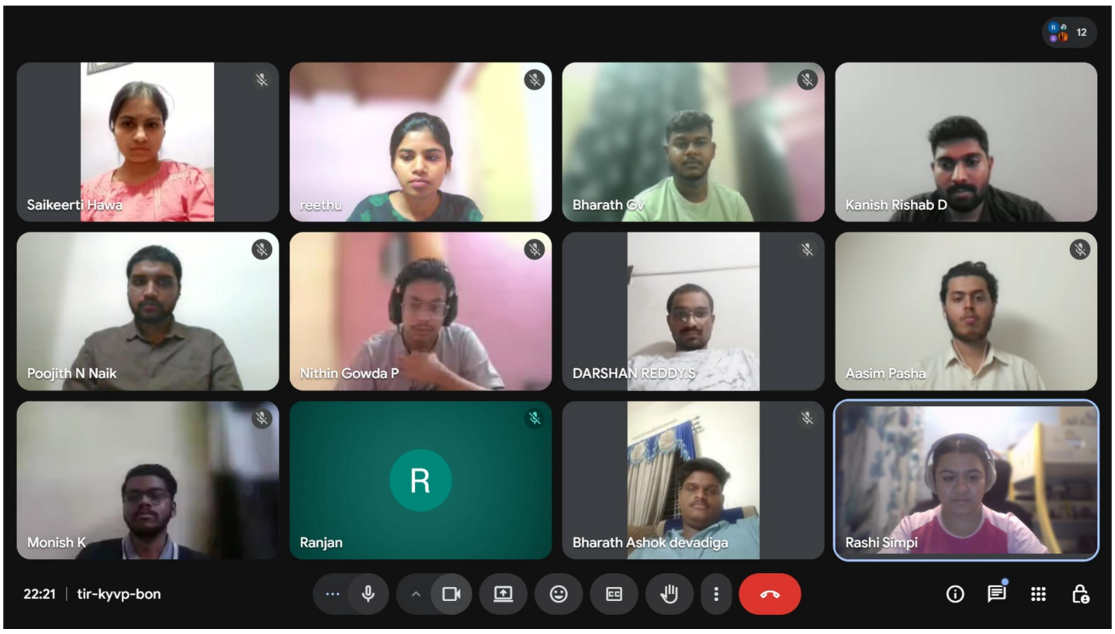
Team Members Absent: 3 (Mohammed Bilal Raza, Mohammed Sammer, Khadija Tul Kubra)

Objective of the day: To evaluate the performance of the trained machine learning models and compare them using different evaluation metrics in order to identify the most reliable model for detecting potential poaching or illegal logging sounds.

Work Completed: The evaluation of the Random Forest classifier was carried out by calculating important performance metrics such as accuracy, precision, recall, and F1-score to measure the effectiveness of the model. Additional metrics including false positive rate and false negative rate were also computed to understand how often the model incorrectly predicts threat events or misses actual threats. A confusion matrix heatmap was generated to visually analyze the prediction results, and a ROC curve along with the AUC score was plotted to evaluate the model's ability to distinguish between different sound categories.

Further evaluation was performed for the K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) models. For both models, performance metrics such as accuracy, precision, recall, and F1-score were calculated to assess classification performance. False positive and false negative rates were also analyzed to determine how frequently the models produce incorrect predictions. Confusion matrix visualizations were created to better understand the classification results, and ROC curves with AUC scores were plotted for both models. These visualizations helped in comparing how effectively each model separates the different sound classes.

To ensure the reliability of the trained models, cross-validation techniques were applied. The models were evaluated using 5-fold cross-validation, where the dataset was divided into five subsets and the models were trained and tested multiple times using different combinations of training and validation data. The accuracy scores obtained from each fold were recorded and averaged to estimate the overall performance and consistency of the models. Finally, the models were compared based on evaluation metrics such as accuracy and false positive rate. From this comparison, the Support Vector Machine (SVM) model achieved the highest accuracy and the lowest false positive rate, making it the most suitable model for the poaching threat detection task.

Doubts/Problems encountered during the Session: During the evaluation process, some challenges were faced while interpreting multiple evaluation metrics and comparing the performance of different models. Understanding the significance of metrics such as false positive rate and ROC-AUC required careful analysis. Additionally, ensuring proper feature scaling and selecting suitable validation methods were important considerations to achieve reliable model performance.

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